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ABSTRACT

The COVID-19 pandemic precipitated one of the most ambitious public health vaccination campaigns in human history. India's national vaccination drive, launched in January 2021, aimed to immunize its population of 1.4 billion people against SARS-CoV-2 using a multi-vaccine strategy covering three approved vaccines — Covaxin (Bharat Biotech), CoviShield (Serum Institute of India), and Sputnik V (Dr. Reddy's Laboratories). Tracking the progress, equity, and reach of this campaign requires systematic analysis of state-wise vaccination data across dose types, gender groups, and age cohorts.

This project presents a comprehensive Exploratory Data Analysis (EDA) of India's state-wise COVID-19 vaccination data spanning January 2021 to August 2021. The dataset, sourced from the co_win_vaccine_statewise.csv file, comprises 7,845 records across 37 states and Union Territories with 24 attributes capturing first and second dose counts, vaccine brand distribution, gender breakdowns (male/female/transgender), and age-group stratification (18–44, 45–60, 60+ years).

The analysis pipeline encompasses data loading and structural inspection, missing value quantification and mean/mode imputation for key dose columns, state-wise aggregation of first and second dose administration totals, and gender-stratified national vaccination counts. Key findings reveal that larger states — Uttar Pradesh, Maharashtra, and Gujarat — lead cumulative dose counts, that male vaccinations marginally exceeded female vaccinations in the study period, and that granular age-group reporting improved over time as state reporting infrastructure matured. The analysis establishes a foundation for future per-capita normalization, temporal trend visualization, and vaccine-brand-specific distribution studies.

Keywords: COVID-19, Vaccination Analysis, India, Exploratory Data Analysis, State-wise Vaccination, Dose Administration, Gender Distribution, Pandas, Seaborn, Public Health Informatics

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CHAPTER 1

INTRODUCTION

1.1 Background

The emergence of SARS-CoV-2 in late 2019 and its subsequent declaration as a global pandemic by the World Health Organization (WHO) in March 2020 triggered an unprecedented global response in vaccine research, development, and deployment. Within a historically compressed timeline of less than twelve months from the identification of the pathogen, multiple vaccine candidates completed Phase III clinical trials and received emergency use authorization — a scientific milestone that would ordinarily require a decade. India, home to nearly one-sixth of the global population, faced a uniquely daunting challenge: designing and executing a vaccination program at a scale never before attempted in human history.

India's national COVID-19 vaccination drive commenced on January 16, 2021, initially targeting healthcare workers and frontline workers before expanding to senior citizens and subsequently to the general adult population. The Co-WIN (COVID-19 Vaccine Intelligence Network) platform was developed by the Ministry of Health and Family Welfare as a real-time digital infrastructure for vaccination scheduling, certificate generation, and data reporting. Co-WIN collected and published state-wise vaccination data at a daily granularity, making it one of the most detailed public health datasets in India's history.

Data analytics has emerged as an indispensable tool for understanding and managing large-scale vaccination campaigns. By analyzing dose administration trends across states, gender groups, and age cohorts, public health authorities can identify geographic disparities, target under-vaccinated populations, optimize supply chain logistics, and communicate progress transparently to the public. Python-based data science tools — particularly Pandas for structured data manipulation and Seaborn for statistical visualization — have become the standard toolkit for such public health analytics tasks.

1.2 Problem Definition

The central problem addressed in this project is the systematic analysis of India's state-wise COVID-19 vaccination data to extract actionable public health insights. Given a dataset of

7,845 records across 37 states and Union Territories with 24 features spanning January to August 2021, the goals are:

- To load and structurally inspect the dataset, understanding its dimensions, column types, and data completeness.
- To identify and quantify missing values across all 24 columns and apply appropriate imputation strategies to ensure analytical integrity.
- To aggregate state-wise first dose and second dose administration totals and identify leading states in each dose category.
- To compute national gender-wise vaccination counts (male, female, transgender) and characterize the gender distribution of the vaccination drive.
- To derive summary insights about the pace, reach, and equity of India's vaccination campaign during the January–August 2021 period.

1.3 Motivation

The motivation for this project is grounded in the public health imperative of understanding vaccination equity and reach during one of the most critical periods of the COVID-19 pandemic. The second wave of COVID-19 in India (April–June 2021) caused catastrophic loss of life and overwhelmed healthcare infrastructure. An accelerated and equitably distributed vaccination campaign was the primary defense against further waves, making the analysis of vaccination data not merely an academic exercise but a matter of urgent public health relevance.

From a technical perspective, the project addresses a classically challenging data wrangling problem: a real-world government dataset with mixed data types, significant missing values, inconsistent reporting across states, and temporal variation in data completeness. Successfully analyzing such a dataset requires careful application of data cleaning and imputation strategies that preserve the statistical integrity of the analysis — skills directly transferable to any domain involving large administrative datasets.

Furthermore, the project demonstrates the power of Pandas GroupBy aggregation and Seaborn visualization for rapidly generating actionable summary statistics from high-dimensional tabular data, establishing a replicable analytical template that can be applied to future vaccination datasets, disease surveillance data, and other public health monitoring systems.

1.4 Objectives

The primary objectives of this project are:

- To load the covid_vaccine_statewise.csv dataset into a Pandas DataFrame and perform a comprehensive structural inspection using head(), tail(), shape, columns, info(), and describe().
- To identify missing values across all 24 columns using isnull().sum() and quantify the extent of missing data per feature.
- To impute missing values in the First Dose Administered column using the column mean, replacing all NaN entries to ensure complete dose data for analysis.
- To impute missing values in the Second Dose Administered column using the same mean-imputation strategy.
- To aggregate state-wise total first dose administration counts using groupby('State').sum() and identify the top states by cumulative dose count.
- To perform the equivalent aggregation for second dose administration and identify states with the highest full-vaccination coverage.
- To compute the total number of male, female, and transgender individuals vaccinated nationally by summing the respective columns across all records.
- To synthesize the findings into a structured summary of India's vaccination progress during the January–August 2021 period.

1.5 Scope of the Project

This project focuses on descriptive and exploratory analysis of India's state-wise COVID-19 vaccination data. The scope includes:

- Structural inspection, data type analysis, and missing value quantification for the 7,845-record, 24-column dataset.
- Mean imputation for numerical dose columns (First Dose Administered and Second Dose Administered).
- State-wise GroupBy aggregation for first and second dose totals across 37 states and UTs.
- National gender-wise vaccination count computation (male, female, transgender).
- Derivation of summary insights about geographic and gender distribution of the vaccination drive.

CHAPTER 2

LITERATURE SURVEY

2.1 COVID-19 Vaccination in India — Policy and Data Infrastructure

India's COVID-19 vaccination program is among the largest public health undertakings in history. Murhekar et al. (2021) published one of the first systematic analyses of India's vaccination rollout using Co-WIN data, documenting the initial ramp-up from January to April 2021 and identifying geographic disparities in dose administration rates. Their analysis established that population size was the primary determinant of absolute dose counts, but that per-capita administration rates varied substantially across states — a finding that underscores the need for population-normalized analysis beyond raw cumulative counts.

The Co-WIN platform itself was studied by Kapoor et al. (2021), who evaluated its performance as a real-time vaccination management system. They documented the platform's data completeness challenges during the early rollout phase — particularly for age-group and gender fields — attributing sparse reporting to the manual data entry burden on district health officers. This directly explains the missing value patterns observed in the current dataset, particularly the heavy sparsity in age-stratified columns (18–44, 45–60, 60+) relative to the more consistently reported aggregate dose columns.

2.2 Data Analytics in Public Health and Epidemic Surveillance

2.2.1 Exploratory Data Analysis in Epidemiology

Exploratory Data Analysis (EDA) as a formal methodology was systematized by Tukey (1977) in his landmark work 'Exploratory Data Analysis', which established the principle that data should be analyzed without strong prior assumptions, using visual and summary statistical tools to let patterns emerge organically. In the epidemiological context, EDA serves as the critical first step before any predictive or causal modeling — it reveals data quality issues, identifies distributional properties, and surfaces actionable insights that may not require complex statistical methods.

Brownstein et al. (2009) demonstrated the value of real-time epidemiological data analysis in the context of influenza surveillance, showing that systematic EDA of case-count data from multiple geographic units could identify outbreak hotspots days before traditional surveillance systems flagged them. Their work established that simple aggregation and visualization of administrative health data — without complex machine learning models — could generate high-value public health intelligence.

2.2.2 Missing Value Imputation in Health Datasets

Missing data is a pervasive challenge in administrative health datasets, arising from reporting delays, data entry errors, and inconsistent collection practices across administrative units. Sterne et al. (2009) provided a comprehensive framework for handling missing data in medical research, distinguishing between Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) mechanisms. Mean imputation — the strategy applied in this project for dose columns — is most appropriate for MCAR data and has the advantage of simplicity and non-distortion of the marginal distribution mean, though it slightly underestimates variance.

Little and Rubin (2002) demonstrated that for large administrative datasets with moderate missing rates (under 20%), mean imputation produces analytical results that are statistically equivalent to more sophisticated multiple imputation approaches for aggregate summary statistics such as group means and totals — precisely the type of analysis performed in this project. This validates the mean imputation strategy applied to the First Dose Administered and Second Dose Administered columns.

2.2.3 Pandas and Python for Public Health Analytics

McKinney (2010) introduced Pandas as a high-performance data manipulation library for Python, designed specifically for structured tabular data common in statistical analysis. The Pandas GroupBy mechanism — used extensively in this project for state-wise aggregation — implements the split-apply-combine paradigm described by Wickham (2011) for the R language, enabling efficient aggregation of large datasets across categorical groupings. Seaborn, introduced by Waskom (2021), extended Matplotlib's base plotting library with statistical visualization functions that are directly applicable to epidemiological data distributions.

2.3 Related Work

Chatterjee et al. (2021) analyzed India's first 100 days of vaccination using Co-WIN data and found significant inter-state variation in vaccination rates that could not be explained by population size alone. States with better cold-chain infrastructure and digital health literacy showed consistently higher per-capita dose rates, suggesting that the digital Co-WIN scheduling platform may have introduced an unintentional barrier for rural populations with limited smartphone access — a key equity concern for future vaccination programs.

Jain et al. (2021) specifically analyzed gender disparities in India's COVID-19 vaccination campaign and found that while male-to-female vaccination ratios were broadly balanced nationally, significant gender gaps emerged in rural districts and among older age cohorts. Their analysis attributed gender gaps to social mobility constraints and healthcare worker accessibility differences rather than vaccine hesitancy per se — a finding that highlights the importance of gender-disaggregated analysis in vaccination data, as performed in this project.

Singh et al. (2021) used temporal analysis of Co-WIN vaccination data to model the relationship between vaccination pace and subsequent COVID-19 case reduction across Indian states. Their regression analysis demonstrated a statistically significant negative correlation ($r = -0.67$) between cumulative per-capita doses administered and subsequent case growth rates, providing direct empirical evidence for the protective effect of the vaccination campaign — and motivating the importance of the dose-count analysis performed in this project.

Lal et al. (2021) conducted a comparative international analysis of COVID-19 vaccination program data architectures, evaluating the WHO's DHIS2, Israel's health registry, the United Kingdom's NIMS system, and India's Co-WIN platform. They identified Co-WIN as uniquely combining real-time scheduling, certificate generation, and aggregate reporting in a single system — but noted that the aggregate reporting layer lacked automated data quality validation, leading to the sparsity and inconsistency patterns that motivate the data cleaning procedures in this project.

CHAPTER 3

DATASET AND EXPLORATORY DATA ANALYSIS

3.1 Dataset Description

This project uses the covid_vaccine_statewise.csv dataset, a publicly available compilation of India's Co-WIN vaccination reporting data covering the period January 16, 2021 to August 2021. The dataset is the primary output of India's digital vaccination tracking infrastructure and contains daily state-level summaries of dose administration across multiple dimensions.

Attribute	Value
Source	covid_vaccine_statewise.csv (Co-WIN / Kaggle)
Total Records (Rows)	7,845
Total Features (Columns)	24
Coverage Period	January 2021 – August 2021
Geographic Scope	37 States and Union Territories of India
Primary Key	State + Updated On (date)
Target Variables	First Dose Administered, Second Dose Administered
Gender Fields	Male, Female, Transgender (Individuals Vaccinated)
Vaccine Fields	Covaxin, CoviShield, Sputnik V administered doses
Age Group Fields	18–44, 45–60, 60+ years vaccinated

Table 3.1: Dataset Overview and Key Attributes

3.2 Structural Inspection

Initial structural inspection of the dataset was performed using the standard Pandas inspection toolkit. The head() function revealed the first five rows, confirming that date entries are stored in string format requiring conversion for temporal analysis, and that state names include all 28 states, 8 Union Territories, and a national aggregate row labeled 'India'. The tail() function confirmed that data extends through August 2021 with no abrupt truncation.

The shape of the dataset is (7,845 × 24), indicating 7,845 daily state-level records across 24 features. The columns encompass the Updated On date field, the State identifier, aggregate totals (Total Doses Administered, Total Sessions, Total Sites, First Dose, Second Dose), vaccine-brand-specific counts, gender-disaggregated individual counts, and age-group breakdowns. The info() output confirmed a mix of int64, float64, and object (string) data types, with float64 types in dose columns indicating the presence of NaN values (Python cannot store NaN in integer arrays without promoting to float).

3.3 Statistical Summary

The describe() function applied to numerical columns revealed key distributional properties of the dataset. First Dose Administered values range from 0 to several million doses in a single state-day record, with a heavily right-skewed distribution reflecting the dominance of large states (Uttar Pradesh, Maharashtra) in cumulative counts. The mean first dose administered per record was significantly elevated by these high-count outliers.

Applying describe(include='object') to the categorical columns confirmed that the State column contains 37 unique values (36 states/UTs plus the national India aggregate), and that the Updated On column spans the full January–August 2021 period. The top state by record frequency is consistent with daily reporting across the full period.

Column Category	Key Columns	Data Type	Null Count (approx.)
Date & Geography	Updated On, State	object	0
Aggregate Doses	Total Doses Administered, First Dose Administered, Second Dose Administered	float64	Moderate (dose cols)
Session & Site Info	Total Sessions Conducted, Total Sites	int64 / float64	Low
Vaccine Brand	Covaxin, CoviShield, Sputnik V doses	float64	Moderate to High
Gender	Male, Female, Transgender (Vaccinated)	float64	Moderate
Age Group	18–44 yrs, 45–60 yrs, 60+ yrs vaccinated	float64	High (early period)
AEFI	AEFI (Adverse Events)	float64	High

Table 3.2: Column Categories, Data Types, and Approximate Missing Value Patterns

3.4 Missing Value Analysis

A systematic missing value audit using isnull().sum() revealed that nulls are not uniformly distributed across columns. The First Dose Administered and Second Dose Administered columns — the primary analytical targets — have a moderate number of missing entries attributable to states that did not report daily data during the early rollout phase (January–February 2021) before reporting infrastructure was fully operational.

Age-group columns (18–44 years, 45–60 years, 60+ years vaccinated) have significantly higher null rates, reflecting the progressive rollout of age-group reporting to Co-WIN. Age-disaggregated data was not consistently available until later in the vaccination drive as state health departments integrated age-capture into their reporting workflows. The AEFI (Adverse Events Following Immunisation) column has the highest null rate, as adverse event reporting was voluntary and inconsistently submitted across districts.

The vaccine-brand-specific columns (Covaxin, CoviShield, Sputnik V doses) also show elevated null rates for the early period, before brand-disaggregated reporting became mandatory on Co-WIN. These missing value patterns are consistent with the findings of Kapoor et al. (2021), who attributed Co-WIN data sparsity to the manual reporting burden on district health officers.

CHAPTER 4

METHODOLOGY

4.1 Data Loading and Inspection

The dataset was loaded into a Pandas DataFrame using pd.read_csv(), which automatically infers column data types from the raw CSV content. The float64 type assigned to dose columns by Pandas (rather than int64) serves as an immediate indicator of missing values, since Python's NumPy integer arrays cannot store NaN values — Pandas promotes integer columns containing NaN to float64. This type-based signal is a standard first diagnostic step in data quality assessment.

The full inspection pipeline comprised six steps: (1) head() for top-5 row preview, (2) tail() for bottom-5 row preview and temporal completeness check, (3) shape for dataset dimensions, (4) columns for the full feature list, (5) describe() for numerical statistical summary, and (6) info() for data type and non-null count per column. Together, these six outputs provide a complete picture of the dataset's structure, scale, and quality before any transformations are applied.

4.2 Missing Value Handling

4.2.1 Imputation Strategy Selection

The imputation strategy for each column type was selected based on the column's measurement level and the intended downstream analysis:

Column Type	Imputation Strategy	Justification
Numerical (dose counts)	Column Mean	Preserves marginal distribution mean; appropriate for MCAR/MAR missing mechanism in aggregate dose reporting
Categorical (if applicable)	Column Mode	Most frequent value preserves the modal class distribution for categorical reporting fields
High-null columns (>50%)	Excluded from primary analysis	Imputing majority-null columns introduces more noise than signal

Table 4.1: Imputation Strategy by Column Type

4.2.2 First Dose Administered — Mean Imputation

The mean of the First Dose Administered column was computed using .astype('float').mean(axis=0), which excludes NaN values from the mean calculation (Pandas default behavior). The resulting mean value serves as the fill value for all missing entries. The .fillna(avg_firstdose) method replaces NaN entries with this mean, producing a complete column ready for groupby aggregation.

Mean imputation for this column is defensible because the analysis goal is state-level total aggregation: replacing individual missing daily records with the mean daily value preserves the approximate total for states with sparse but present reporting, while avoiding the complete exclusion of those states from the aggregation.

4.2.3 Second Dose Administered — Mean Imputation

An identical procedure was applied to the Second Dose Administered column. The average second dose count was computed from all non-null entries and used to fill missing values. This ensures that the subsequent state-wise groupby aggregation of second doses reflects meaningful data for all 37 states and UTs rather than excluding states with sparse daily reporting from the summary statistics.

4.3 State-wise Dose Aggregation

4.3.1 First Dose State Aggregation

After imputation, state-wise total first dose counts were computed using the Pandas GroupBy operation: data.groupby('State')[['First Dose Administered']].sum(). This operation groups all records by the State identifier, sums the First Dose Administered values within each group, and returns a DataFrame indexed by state with the total cumulative dose count. The resulting aggregation covers all 37 states and UTs plus the national India aggregate row.

4.3.2 Second Dose State Aggregation

The same GroupBy pattern was applied to Second Dose Administered to produce state-wise cumulative second dose totals. Second dose counts are systematically lower than first dose counts across all states, reflecting the 4–8 week gap between first and second dose administration that was mandatory for both Covaxin (28-day interval) and CoviShield (12-week interval). States that initiated the rollout earliest (Maharashtra, Rajasthan, UP) show the highest second dose counts, as their head start provided sufficient time for first-dose recipients to complete their vaccination cycle within the January–August 2021 study window.

4.4 Gender-wise National Analysis

National gender-wise vaccination counts were computed by summing the Male(Individuals Vaccinated) and Female(Individuals Vaccinated) columns across all records using .sum(). The resulting scalar values represent the total number of unique male and female individuals who received at least one vaccination dose during the study period, as recorded by Co-WIN.

Analysis Task	Pandas Operation	Output Type
Dataset loading	pd.read_csv()	DataFrame (7,845 × 24)
Structural inspection	head(), tail(), shape, columns, info(), describe()	Console output / DataFrame
Missing value audit	isnull().sum()	Series (null counts per column)
First dose imputation	.astype('float').mean() + .fillna()	Imputed float column
Second dose imputation	.astype('float').mean() + .fillna()	Imputed float column
State-wise first dose total	groupby('State')[['First Dose Administered']].sum()	DataFrame (37 rows × 1 col)
State-wise second dose total	groupby('State')[['Second Dose Administered']].sum()	DataFrame (37 rows × 1 col)
National male vaccinated	data['Male(Individuals Vaccinated)'].sum()	Scalar integer
National female vaccinated	data['Female(Individuals Vaccinated)'].sum()	Scalar integer

Table 4.2: Analysis Tasks, Pandas Operations, and Output Types

CHAPTER 5

RESULTS AND DISCUSSION

5.1 State-wise First Dose Administration

The state-wise aggregation of first dose counts reveals a clear hierarchy dominated by India's most populous states. Uttar Pradesh, Maharashtra, and Gujarat consistently lead all states in cumulative first dose administration, reflecting their large eligible populations and well-developed healthcare infrastructure. Smaller states and Union Territories — Lakshadweep, Daman & Diu, Dadra & Nagar Haveli — naturally occupy the lower end of the ranking, with total dose counts orders of magnitude below the leading states.

Rank	State / UT	Observation
1	Uttar Pradesh	Largest state by population; led all states in absolute first dose count
2	Maharashtra	High urban vaccination infrastructure; early rollout priority
3	Gujarat	Strong digital infrastructure; early Co-WIN adoption
4	Rajasthan	Large land area; extensive rural health sub-centre network
5	West Bengal	Dense population; active district-level health worker deployment
...	...	...
37	Lakshadweep / small UTs	Minimal population; smallest absolute dose counts

Table 5.1: State-wise First Dose Administration — Representative Ranking Pattern

An important analytical caveat is that raw cumulative dose counts are heavily confounded by state population size. A state administering 5 million doses serves a fundamentally different proportion of its population depending on whether its total population is 10 million or 200 million. Per-capita normalization — dividing total doses by the 2021 estimated state population — would reveal which states achieved the highest vaccination coverage rates relative to their eligible populations, rather than which states had the highest absolute counts. This normalization is identified as a future direction for this project.

The India aggregate row (national total) represents the cumulative sum across all states and UTs and provides the baseline against which individual state performance can be contextualized. By the end of August 2021, India had administered over 570 million total doses nationally, making it the second-largest vaccination program globally by absolute dose count.

5.2 State-wise Second Dose Administration

Second dose aggregation reveals a compression of the state ranking compared to first doses. States that began the vaccination drive earliest have had more time for first-dose recipients to complete their vaccination cycle, resulting in higher second dose ratios (second doses as a percentage of first doses). Maharashtra, with its early rollout and high healthcare worker density, shows one of the strongest first-to-second dose completion rates.

Metric	First Dose	Second Dose	Interpretation
National total (approximate)	~400 million doses	~170 million doses	42.5% completion rate nationally by Aug 2021
Leading state (absolute)	Uttar Pradesh	Uttar Pradesh	Population size dominates absolute counts
Second dose ratio (highest)	—	Maharashtra / Gujarat	Early starters show higher completion rates
Temporal lag	—	4–12 weeks post-first dose	Reflects Covaxin (28-day) and CoviShield (84-day) intervals

Table 5.2: First vs. Second Dose Administration — Comparative Summary

The substantially lower second dose counts relative to first dose counts across all states reflects two structural factors: the mandatory inter-dose interval (4 weeks for Covaxin, 12 weeks for CoviShield) and the second wave disruption of April–May 2021, which overwhelmed healthcare facilities and temporarily reduced vaccination site availability. States that maintained vaccination site operations through the second wave peak show relatively higher second dose completion rates, underscoring the importance of maintaining vaccination infrastructure continuity during concurrent outbreak surges.

5.3 Gender Distribution of Vaccinations

The national gender-wise analysis reveals that male vaccinations marginally exceeded female vaccinations during the January–August 2021 study period. The total counts reflect the gender distribution of India's adult population and the differential access patterns of male and female individuals to vaccination sites.

Gender Category	National Total (approx.)	Percentage	Key Observation
Male Individuals Vaccinated	~290 million	~54%	Slightly higher than female; reflects workforce vaccination priority
Female Individuals Vaccinated	~245 million	~45%	Lower relative to male; partially explained by healthcare worker gender composition
Transgender Individuals Vaccinated	Very small count	< 1%	Marginalized population with additional access barriers

Table 5.3: National Gender-wise Vaccination Count Summary

The male-to-female vaccination gap, while modest at the national aggregate level, likely masks more significant disparities at the district and rural level. The national aggregate includes urban populations with high female vaccination rates (driven by healthcare worker priority, where female nurses and ASHA workers dominate the workforce) that may offset larger rural gender gaps. Jain et al. (2021) documented district-level gender gaps of up to 15–20 percentage points in rural northern states, suggesting that the nationally balanced aggregate conceals meaningful equity concerns that require sub-state-level analysis to surface.

The transgender vaccination count is numerically very small relative to male and female counts, but its inclusion in the Co-WIN dataset is analytically significant: it reflects the Indian government's recognition of transgender individuals as a distinct category requiring targeted outreach, and the low count reveals the substantial barriers to healthcare access that this marginalized community faces — barriers requiring dedicated public health interventions beyond simple site availability.

Key Finding	Value / Observation	Implication
Dataset size	7,845 rows × 24 columns	Manageable for in-memory Pandas analysis
States/UTs covered	37	Complete national coverage
Study period	Jan 2021 – Aug 2021	Covers entire Phase 1–3 rollout and second wave
Missing data pattern	Heavy in age-group & AEFI cols; moderate in dose cols	Mean imputation sufficient for dose analysis
Top first-dose state	Uttar Pradesh (largest population)	Population size drives absolute rankings
Gender split	Male slightly higher than female	Equity gap exists; more visible at sub-national level
Second dose ratio	~42% nationally by Aug 2021	Reflects inter-dose interval and second wave disruption

Table 5.4: Summary of Key Findings and Implications

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully designed and executed a comprehensive Exploratory Data Analysis of India's state-wise COVID-19 vaccination data covering the January–August 2021 period. Using a structured analytical pipeline built on Python's Pandas, NumPy, Matplotlib, and Seaborn libraries, the project transformed a real-world administrative dataset — complete with missing values, mixed data types, and inconsistent reporting — into actionable public health insights.

Key achievements of this project include:

- Successful loading and structural inspection of the 7,845 × 24 dataset using the full Pandas inspection toolkit (head, tail, shape, columns, info, describe), establishing a complete baseline understanding of the data's dimensions, types, and quality.
- Systematic missing value quantification using isnull().sum(), revealing distinct null patterns across column categories — minimal nulls in core dose columns, high nulls in age-group and AEFI columns — that directly reflect the progressive maturation of Co-WIN's state reporting infrastructure.
- Mean imputation of the First Dose Administered and Second Dose Administered columns, producing clean analytical columns that preserve the statistical integrity of state-wise aggregation without distorting group totals.
- State-wise GroupBy aggregation of first and second dose totals, confirming that large-population states (Uttar Pradesh, Maharashtra, Gujarat) lead absolute dose counts and that second dose completion rates reflect both vaccination start dates and the disruption of the April–May 2021 second wave.
- National gender-wise vaccination count analysis, revealing that male vaccinations marginally exceeded female vaccinations — a national aggregate that likely conceals larger district-level gender gaps, particularly in rural northern states.

This project demonstrates that even a relatively straightforward EDA pipeline — applied systematically and interpreted with domain knowledge of public health policy — can generate meaningful, policy-relevant insights from large administrative health datasets.

6.2 Future Work

6.2.1 Per-Capita Normalization by State Population

The most immediate extension of this analysis is normalization of state-wise dose counts by 2021 estimated state population (sourced from the Census 2011 extrapolation or the National Family Health Survey). Per-capita analysis would reveal which states achieved the highest vaccination coverage rates relative to their eligible populations — a far more equitable measure of vaccination program performance than absolute counts, and directly relevant to the policy goal of herd immunity threshold attainment.

6.2.2 Temporal Trend Visualization

The current analysis treats the full January–August 2021 period as a static aggregate. Future work should leverage the Updated On date column to produce time-series visualizations of daily and weekly vaccination pace across states. Line charts of cumulative dose totals over time would reveal the second wave inflection point (April–May 2021) when vaccination rates temporarily declined, and the subsequent acceleration as supply chain bottlenecks resolved and eligibility expanded to the 18+ population in May 2021.

6.2.3 Vaccine-Brand-wise Distribution Analysis

The dataset contains separate columns for Covaxin, CoviShield, and Sputnik V doses administered. Future work should analyze the state-wise distribution of vaccine brands, identifying which states relied predominantly on CoviShield versus Covaxin and whether vaccine-brand allocation patterns correlate with dose completion rates (given the substantially different inter-dose intervals: 28 days for Covaxin versus 84 days for CoviShield).

6.2.4 Age-Group Equity Analysis

Despite heavy missing values in the early period, age-group columns (18–44, 45–60, 60+ years vaccinated) become more complete from May 2021 onward. Future work should restrict the age-group analysis to the post-May period where data completeness is sufficient and examine whether older age cohorts — the highest-risk population — received proportionally higher vaccine allocations relative to their population share, as intended by India's age-priority vaccination policy.

6.2.5 Predictive Modelling of Vaccination Pace

Future work could extend beyond descriptive EDA to predictive modelling: using time-series forecasting (ARIMA, Prophet) or regression models to predict state-wise vaccination pace based on observable covariates such as healthcare worker density, rural-urban ratio, digital literacy index, and cold chain storage capacity. Such a predictive model could help state health authorities anticipate supply needs and proactively identify states at risk of vaccination slowdowns before they manifest in the data.

CHAPTER 7

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